

DAGF: A dual GCN and auxiliary graph fusion based model for aspect-based sentiment analysis

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HIGHLIGHTS

- Solving the ABSA problem via dual GCN and auxiliary graph fusion techniques.
- Capturing syntactic structure via knowledge-enhanced syntactic GCN and relation-aware self-attention.
- Utilizing AMR to construct semantic GCN and employing KL divergence to measure learning discrepancy within dual GCN.
- Auxiliary graph fusion integrates dual GCN for collaborative learning between syntactic and semantic representations.

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ABSTRACT

Aspect-based Sentiment Analysis (ABSA) is a fine-grained sentiment analysis task that aims to distinguish the sentiment polarity corresponding to different aspects in a sentence. Traditional solutions have utilized syntactic and semantic information but face two core challenges: First, how to establish robust dependencies between aspect words and their corresponding contexts in syntactic analysis. Second, how to better capture deep semantic relations between aspect words and opinion words. To address these challenges, we propose the Dual Graph Convolutional Network and Auxiliary Graph Fusion (DAGF) model. The model constructs a syntactic Graph Convolutional Network (GCN) and a semantic GCN to improve syntactic and semantic analyses. An auxiliary graph fusion module is designed to capture the complex connections between syntactic and semantic structures. In particular, our approach strengthens dependency modeling in the syntactic GCN through the fusion of an external knowledge base with dependency trees and the introduction of a relation-aware self-attention mechanism. For deeper semantic capture, we implement Abstract Meaning Representation (AMR) within the semantic GCN, replacing traditional semantic representations. We also develop an auxiliary graph fusion module that integrates the syntactic and semantic GCNs, facilitating collaborative learning across both representation types. Experiments on three public datasets show that DAGF achieves the highest F1 scores, improving by 1.92, 1.54 and 0.15, respectively. It also attains the best accuracy on the first two datasets and second-best on the third, with accuracy gains of 1.61, 1.25 and -0.59 over the strongest baselines.

1. Introduction

Sentiment Analysis (SA) is one of the key research topics in Natural Language Processing (NLP) [1]. It aims to determine the overall sentiment polarity of a text and is commonly used on various types of social networks to analyze user attitudes. With the rapid development of the Internet, corpora featuring more complex semantics and ambiguous expressions have emerged. To address these issues, research on Aspect-Based Sentiment Analysis (ABSA) has emerged. It is an

entity-specific and more fine-grained sentiment analysis task that aims to determine the sentiment polarity (e.g., positive, negative, or neutral) of aspects appearing in a text. For example, consider the restaurant review in Fig. 1: “The food is delicious but the service is terrible.” In this sentence, the two aspect terms are food and service, and their corresponding sentiment polarities are “positive” and “negative”, respectively. Unlike sentence-based sentiment analysis, ABSA can accurately determine the sentiment polarity of each specific aspect.

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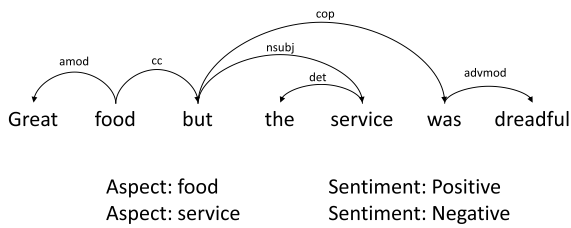


Fig. 1. An example of the ABSA problem.

A primary challenge in ABSA lies in accurately modeling the dependency relationships between aspect words and their corresponding contexts, especially with opinion words. This problem becomes more difficult when sentences contain multiple aspects and complex opinion expressions. Traditional methods are based on neural networks with attention mechanisms. For example, Yang et al. [2] proposed a hierarchical attention network that gradually focuses on key information related to the aspect words through a multi-layer attention mechanism. Yadav et al. [3] proposed an attention-based, position-independent approach for ABSA that uses polarity substitution and masking techniques to preserve aspect word information. Wang et al. [4] developed an attention-enhanced Long Short-Term Memory (ATAE-LSTM) model for ABSA, specifically improving contextual focus around aspect-related words. Chen et al. [5] introduced a semantic information extraction strategy based on memory rotation units to obtain long-term semantic information in context and construct the memory network. Gu et al. [6] proposed a location-aware bidirectional attention network that enables the model to better distinguish the effects of words in different positions on aspect words by introducing location information. However, sentential noise can interfere with the model's ability to capture the relationships between aspect words and opinion words.

Since Graph Neural Networks (GNNs) have better structural perception, applying GNNs constructed from syntactic dependencies to ABSA tasks yields better results compared to traditional neural networks. Recent studies [7,8] have proposed GCNs and graph attention networks (GATs) based on syntactic dependency trees to help the attention mechanism better establish the connection between aspects and context. However, dependency tree-based GCNs often exhibit lower accuracy on datasets where sentences contain weak syntactic signals but rich semantic information. This is mainly because such models struggle to distinguish semantically similar syntactic structures and are susceptible to noise introduced by dependency parsing. More critically, when aspect and opinion words share identical syntactic relationships but express different sentiments (e.g., “Great food but the service was dreadful.”), syntax-based models may fail to discern the contrast due to their reliance on structural patterns, overlooking pivotal semantic cues like the conjunction “but.” Additionally, implicit opinion expressions that lack direct syntactic links to aspects (e.g., “The restaurant was a disappointment” expressing negative sentiment toward multiple aspects) cannot be effectively captured through dependency paths alone.

Another challenge in ABSA lies in effectively exploring and utilizing semantic information to establish connections between aspect words and opinion words. For example, in Fig. 1, the conjunction “but” may influence the judgment of the sentiment polarity of “food.” Modeling semantic information often faces issues of unclear or loose association paths. To address this challenge, Ma et al. [9] proposed a model that abandons syntactic structures in favor of deeper semantic representations. This approach enables more flexible and comprehensive use of semantic information to identify potential connections between aspects and opinion words. However, while this semantic structure aids in understanding semantic links between words, syntactic dependency structures retain greater advantages in modeling long-distance

dependencies in longer sentences. Specifically, in complex sentences with multiple clauses, purely semantic models struggle to determine which opinion words correspond to which aspects when they are separated by significant textual distance, as semantic similarity alone provides insufficient structural guidance. For example, in “Although the ambiance was charming, which I appreciated, the food, particularly the main course, was disappointing,” semantic approaches may incorrectly associate “appreciated” with “food” due to semantic proximity, whereas syntactic structure clearly delineates the correct aspect–opinion pairings through grammatical dependencies. Therefore, effectively integrating syntactic and semantic information remains a key challenge in ABSA research.

To solve the above challenges, this paper proposes the DAGF model to integrate syntactic and semantic information in sentences. The model incorporates a dual GCN structure comprising a syntactic GCN and a semantic GCN. In syntactic GCN, we propose a method that combines knowledge enhancement with syntactic dependencies. Initially, we use the probability matrix of all dependency edges from the dependency parser to model the syntactic GCN. This matrix typically contains rich syntactic information. Considering that sentences include non-opinion words, we integrate an opinion lexicon containing common opinion and non-opinion words categorized by sentiment polarity. We then perform secondary modeling with aspects and sentiment polarity to enhance the contextual classification ability of aspect terms. In semantic GCN, AMR is introduced to construct the semantic GCN, providing a robust semantic structure. As shown in Fig. 1, “service” and “dreadful” are directly connected semantically, ignoring function words such as “but” and “was”, which facilitate establishing links between aspects and opinion words. Additionally, we incorporate Kullback-Leibler (KL) divergence loss to quantify the error in feature learning by the two GCNs. Furthermore, we designed a simple and effective auxiliary graph fusion module to fuse syntactic features and semantic features. In summary, our main contributions are as follows:

- We propose a novel DAGF model that utilizes a dual GCN and an auxiliary graph fusion module to solve the ABSA problem.
- We propose a knowledge-enhanced syntactic GCN that leverages external sentiment lexicons to establish robust polarity connections, coupled with relation-aware self-attention that captures structural information while reducing the unreliability of traditional parsers.
- We advance semantic representation by incorporating AMR into the semantic GCN, with a KL divergence-based regularization mechanism ensuring semantic consistency between learned features and semantic structures.
- We design a dual-channel auxiliary graph fusion module that effectively integrates syntactic and semantic GCNs, resulting in more collaborative learning between syntactic and semantic representations.
- Extensive experiments on three benchmark datasets demonstrate the effectiveness of the DAGF model, achieving superior performance compared to state-of-the-art methods.

The rest of the paper is structured as follows: Section 2 introduces the related work. Section 3 details the DAGF model. Experiments and analyses are presented in Sections 4 and 5 concludes the paper. Our code is available at <https://github.com/zwl1985/DAGF>.

2. Related work

ABSA provides a more fine-grained perspective than traditional sentiment analysis, which typically focuses on sentence or document-level sentiment. By operating at the entity level, ABSA enables more granular sentiment interpretation. Earlier studies about ABSA [10–12] primarily relied on manually extracted aspect features, which limited their ability to establish dependencies between a given aspect and its contextual information.

Recently, Wang et al. [4], Ma et al. [13] and Fan et al. [14] used attention-based neural networks to implicitly model the dependencies between aspect words and context. For example, Wang et al. [4] proposed ATA-E-LSTM model applied to the ABSA task. Similarly, Chen et al. [5] and Chen et al. [15] both utilized a hierarchical attention network to identify the sentiment polarity of a given aspect. Fan et al. [14] captured word-based interactions between aspects and contexts using a dual attention neural network with a multi-granularity attention mechanism. Additionally, the pre-trained language model Bidirectional Encoder Representations from Transformers (BERT) [16] achieved notable performance in the ABSA task. Xu et al. [17] proposed a post-training method for BERT to enhance performance during the fine-tuning phase.

Syntactic analysis and semantic analysis are two primary research components in the ABSA task. In syntactic analysis, a dominant approach involves learning syntactic perceptual features using syntactic dependency trees to facilitate connections between aspect words and other words. For example, Sun et al. [18] employed a GCN layer to extract syntactic information from the dependency tree. Similarly, Liang et al. [19] developed aspect-specific sentiment features by constructing a unified graph structure centered on aspect words to capture relationships among these words. Additionally, Zhang et al. [20] created a global lexical graph to capture word co-occurrence relations and integrated it with a syntactic graph. Liang et al. [21] built an emotion enhancement graph by incorporating external emotional knowledge, focusing on the emotional connections between opinion words and aspect words.

Semantic analyses are often combined with syntactic analyses to jointly extract relationships between aspect words and opinion words. For example, Zhang et al. [22] improved BERT's performance in ABSA through a dynamic semantic reweighting mechanism. Yan et al. [23] proposed a semantically enhanced dual-channel graph communication network that integrates semantic information and syntactic features to generate word representations through communication mechanisms and hierarchical attention modules. Chen et al. [24] defined ten types of word relationships and employed a dual-affine attention module to encode these relationships as adjacency tensors between words. This approach constructs a multi-channel graph structure to learn relationship-aware node representations. Xin et al. [25] enhanced the performance of ABSA by integrating syntactic and semantic information through a hybrid neural network. Ma et al. [9] discarded combining syntactic information and emphasized the use of a more concise and efficient AMR to construct a robust semantic structure graph for extracting sentence feature information.

However, while some of these approaches attempt to combine syntactic and semantic information, they often rely on simple feature concatenation or parallel processing, which fails to achieve a deep and effective structural fusion between these two distinct types of knowledge to obtain richer corpus information. Recent research has explored dual-graph architectures to leverage both semantic and syntactic information for ABSA. For example, Wang et al. [26] proposed an Adaptive Dual Graph Convolution Fusion Network that employs two parallel GCNs: one for learning semantic correlations through attention mechanisms and another for capturing syntactic structures via dependency parsing. To effectively integrate these two information sources, they designed a cross-fusion module using double random similarity matrices to obtain syntactic features in semantic space and semantic features in syntactic space. Zhang et al. [27] developed a knowledge-enhanced dual-channel graph convolutional network that constructs semantic-based and syntactic-based GCNs simultaneously. Their approach enhances sentence semantics with commonsense knowledge and employs multi-head attention to construct semantic graphs while pruning dependency trees to focus on aspect-relevant syntactic information. These dual-graph methods demonstrate the potential of parallel processing of semantic and syntactic information, yet they often process the two channels independently before fusion, which may not fully capture the intricate interdependencies between semantic meanings and syntactic

structures. Specifically, they do not sufficiently explore how to mutually enhance the syntactic structure using semantic context. In this paper, we propose an auxiliary graph fusion module that integrates syntactic and semantic features, thereby enhancing the model's ability to enable aspect terms to more accurately capture contextual information.

2.1. Preliminaries

2.1.1. RoBERTa

The Robustly Optimized BERT Pretraining Approach (RoBERTa) model [28], developed by Facebook, is an enhanced version of the BERT model. It significantly improves semantic representation during the pre-training phase by eliminating the Next Sentence Prediction (NSP) task, utilizing larger training datasets, and extending training duration. As a pre-trained language model based on the Transformer architecture, RoBERTa deeply models contextual information in text through a multi-layered bidirectional attention mechanism. Unlike traditional models that depend on sentence structure, RoBERTa does not explicitly rely on syntactic or semantic structures but learns implicit word relationships through large-scale corpus pre-training to capture long-distance contextual dependencies. In text representation, RoBERTa provides semantically rich, context-embedded representations for each word. This offers a robust semantic foundation for sentiment analysis as one of its downstream tasks.

In this paper, RoBERTa is utilized as an encoder to extract deep semantic representations of each word in the input text, providing effective feature input for the classifier. Compared to structure-dependent models such as GCN and LSTM, RoBERTa effectively models global sentence information without requiring explicit sentence structure specifications.

2.1.2. GCN

Inspired by traditional convolutional neural networks (CNNs) and graph embeddings, GCNs serve as an efficient CNN variant that operates directly on graph structures. GCNs encode local information by performing convolutional operations on directly connected nodes. Given a graph with n nodes, it can be represented by an adjacency matrix $A \in R^{n \times n}$. Recent studies [7,18] have extended the GCN model by encoding dependency trees and incorporating dependency paths between words, constructing the adjacency matrix A based on the syntactic dependencies of the sentence. In the adjacency matrix A , A_{ij} indicates whether the i -th node is connected to the j -th node. In the context of a sentence, this indicates whether the i -th word has a dependency relationship with the j -th word. Specifically, in the adjacency matrix A , if the i -th node is connected to the j -th node, then $A_{ij} = 1$; otherwise, $A_{ij} = 0$. The adjacency matrix A , which consists of 0s and 1s, can be considered the final output of the dependency parser. For each node i in matrix A , the update formula for its hidden state representation (denoted by h_i^l) is as follows.

$$h_i^l = \sigma \left(\sum_{j=1}^n A_{ij} W^l h_j^{l-1} + b^l \right) \quad (1)$$

The node feature matrix of layer l is denoted by h^l , where each row represents the feature vector of a node. The weight matrix is denoted by W^l , the bias term by b^l , and the activation function (e.g., ReLU) by σ .

3. Proposed model

The overall framework of the DAGF model is shown in Fig. 2. The model comprises a word embedding module, a syntactic GCN module (SynGCN), a semantic GCN module (SemGCN), and an auxiliary graph fusion module. In the ABSA task, given a sentence-aspect pair, the word embedding module employs RoBERTa to extract the hidden contextual representation of the sentence. This implicit representation is fed into the SynGCN and SemGCN modules, respectively. The outputs from these two GCN modules undergo interactive processing through the auxiliary

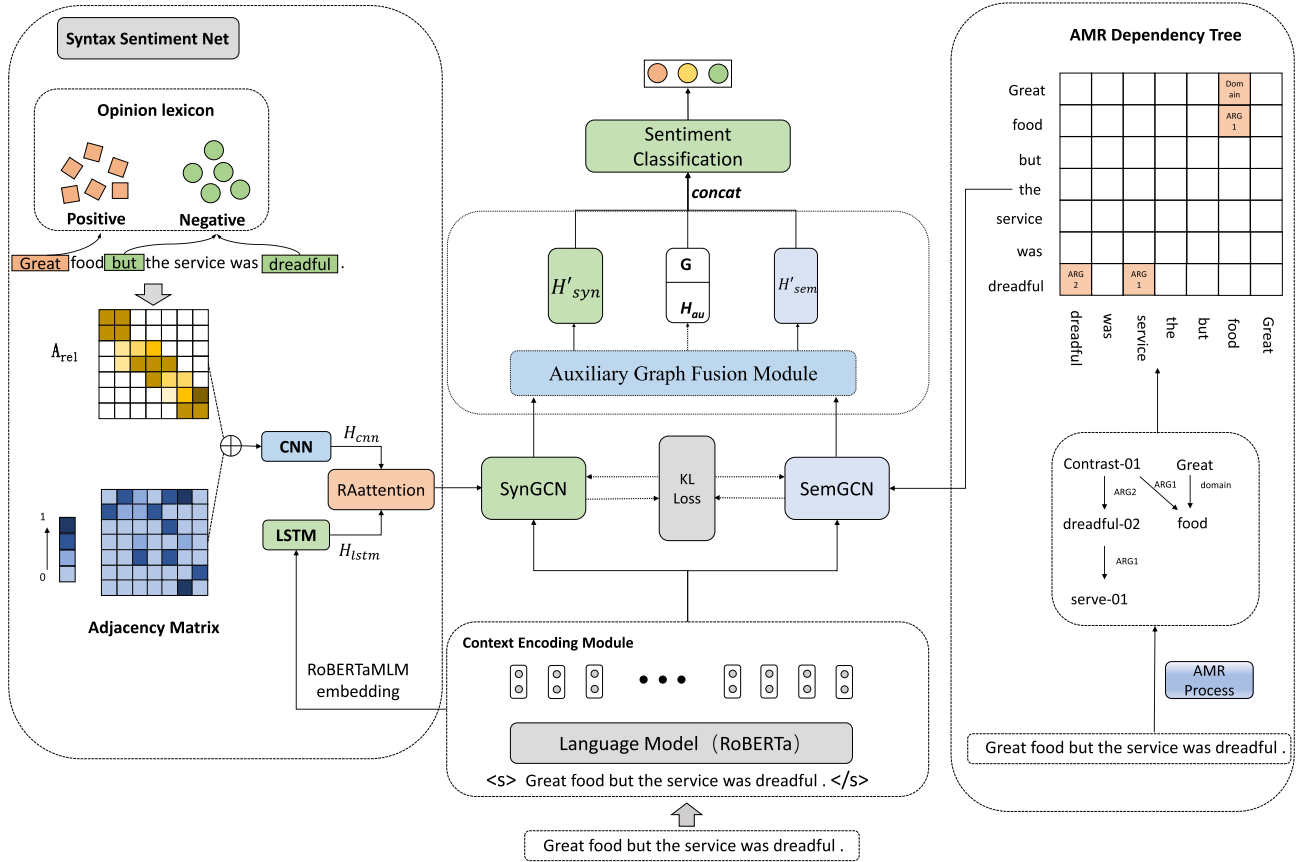


Fig. 2. Overall architecture of the DAGF model. The input sentence is processed through two parallel pathways: the syntactic GCN leverages dependency parsing and external opinion lexicons to capture syntactic dependencies, while the semantic GCN utilizes AMR to model semantic relations. The auxiliary graph fusion module integrates these dual-graph representations to produce the final sentiment prediction for the given aspect term.

graph fusion module to facilitate effective information exchange. The resulting output is inputted into the sentiment classifier for classification. The details of the model are described below.

3.1. Word embedding module

Given a sentence-aspect pair (W, A) , where $W = \{w_1, w_2, \dots, w_r\}$ represents a sentence containing r words and $A = \{a_1, a_2, \dots, a_m\}$ represents an aspect containing m words, the word embedding module first constructs the input text sequence O_{in} for RoBERTa. Using the downstream task of Masked Language Modeling (MLM), it obtains an implicit representation of the sentence-aspect pair. The text sequence O_{in} is constructed as follows.

$$O_{in} = \langle s \rangle W \langle /s \rangle A \langle /s \rangle \quad (2)$$

The input text sequence O_{in} is processed by RoBERTa to obtain the contextual representation of the sentence. To fully leverage different pre-trained models, this paper employs both standard RoBERTa and RoBERTaMLM with MLM encoders to generate distinct contextual representations. The hidden state vectors W_{in} and $W_{in'}$ are obtained as follows.

$$W_{in} = \text{RoBERTa}_{\text{MLM}}(O_{in}) \quad (3)$$

$$W_{in'} = \text{RoBERTa}(O_{in}) \quad (4)$$

where $W_{in} \in \mathbb{R}^{d_{mi} \times n}$, $W_{in'} \in \mathbb{R}^{d_i \times n}$, d_{mi} and d_i are the hidden state dimensions of RoBERTaMLM and RoBERTa, respectively, and n is the length of the input sequence O_{in} .

3.2. SynGCN

The SynGCN module fuses syntactic information and external knowledge for enhanced syntactic structures. First, we take the output W_{in} from the RoBERTaMLM encoder and obtain the sentence's syntactic dependency matrix through the LAL-Parser [29], denoted as $A^{adj} \in \mathbb{R}^{n \times n}$. Then, we construct the opinion lexicon-based adjacency matrix $A^{rel} \in \mathbb{R}^{n \times n}$ based on A^{adj} and the external opinion lexicon O . Specifically, an edge between nodes i and j is established in A^{rel} if they are connected in A^{adj} and at least one of them belongs to O . Next, we concatenate the opinion lexicon-based adjacency matrix A^{rel} with the syntactic dependency matrix A^{adj} to obtain a new matrix A_n . This fusion combines sentiment information from opinion words with syntactic structural information, strengthening the connections between non-opinion words and aspect words while preserving the syntactic relations. Finally, the representation A_n is refined via a relation-aware self-attention mechanism to produce the final output A^{syn} of the SynGCN module.

3.2.1. Opinion lexicon

The opinion lexicon is a vital external resource for sentiment analysis, as it labels the sentiment polarity of commonly used opinion words. Opinion words are crucial for guiding the model in making accurate predictions. We query the sentiment polarity of non-opinion words using Bing Liu's opinion lexicon [30]. As shown in Fig. 3, we find that "but" is a negative non-opinion word and add an opinion relationship between "service" and "but" for the sentence given in Fig. 1.

Unlike constructing a syntactic adjacency matrix, we construct the adjacency matrix A^{rel} by modeling the syntactic dependency tree of the

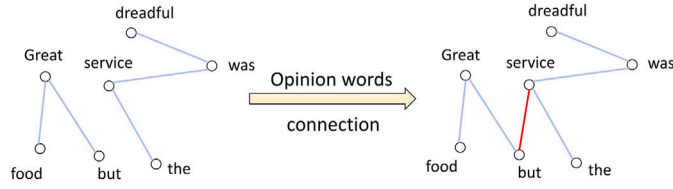


Fig. 3. Illustration of opinion lexicon-based relation enhancement for the sentence “The food is delicious but the service is terrible” (aspect term: ‘service’). In the original syntactic dependency tree, the conjunction ‘but’ and the aspect term ‘service’ have no direct syntactic connection. By querying Bing Liu’s opinion lexicon and identifying ‘but’ as a negative sentiment indicator, we establish a new opinion-based edge between ‘but’ and ‘service’, enabling the model to capture the sentiment contrast that is not explicitly reflected in the syntactic structure.

opinion lexicon and sentences, as shown below.

$$A_{ij}^{rel} = \begin{cases} 1, & \text{if } A_{ij} = 1 \text{ and } (i \in O \text{ or } j \in O) \\ 0, & \text{otherwise} \end{cases} \quad (5)$$

where the adjacency matrix A_{ij} represents the syntactic dependencies within the sentence, i and j denote two nodes connected in the dependency tree, and O represents the external opinion lexicon.

This approach involves first obtaining the dependency tree with syntactic relations from the sentence’s syntactic dependency tree. Then, a second dependency matrix is constructed based on whether two words with syntactic dependency relations appear in the opinion lexicon. If at least one of the two related words is in the opinion lexicon, a relation is established between them; otherwise, no relation is established. This process results in a 0–1 adjacency matrix.

3.2.2. Relation-aware self-attention

To effectively integrate information from the opinion lexicon and syntactic structures, this paper proposes a relation-aware self-attention module. This module processes the fused adjacency matrix and context embeddings through a hybrid neural network approach to generate the final representation of SynGCN. The workflow of the module is as follows.

First, a 2D convolution operation is performed on the fused matrix A_n to generate the initial input for the attention module. Specifically, we design a multi-channel CNN to obtain the feature vector $H_{cnn} \in R^d$ for the CNN component through the ReLU activation function and a max-pooling layer.

Concurrently, we input the RoBERTaMLM embedding vectors from the word embedding module into a unidirectional LSTM network to capture the long-term contextual dependencies of the aspect words, yielding the feature vector output $H_{lstm} \in R^{d^l \times d}$ for the LSTM component, where d^l is the LSTM hidden state dimension.

Finally, the outputs from the CNN and LSTM components are aligned dimensionally and used as inputs to our relation-aware self-attention mechanism for attention computation. The traditional self-attention mechanism [31] is shown as follows.

$$Attention = \text{softmax} \left(\frac{QW^Q \times (KW^K)^T}{\sqrt{d}} \right) \quad (6)$$

where Q and K are d -dimensional input vectors, and W^Q and W^K are learnable weight matrices, both of dimension $R^{d \times d}$.

On this basis, the function of the relation-aware self-attention is as follows.

$$RAttention(Q, K, V) = \text{softmax} \left(\frac{QK^T + R}{\sqrt{d_k}} \right) V \quad (7)$$

where R represents the relationship perception bias matrix, calculated as follows.

$$R = A^{syn} + A^{rel} \quad (8)$$

Based on the above definition, the complete calculation process of the relationship-aware self-attention mechanism is as follows.

$$Att_c = RAttention(Q = A, K = H_{cnn}, V = H_{cnn}) \quad (9)$$

$$Att_l = RAttention(Q = A, K = H_{lstm}, V = H_{lstm}) \quad (10)$$

$$Att_{concat} = \text{concat}(Att_c, Att_l) \quad (11)$$

$$Att = \frac{1}{L} \sum_{i=1}^L Att_{concat,i} \quad (12)$$

where A denotes the aspect word embedding representation obtained after processing the aspect word through the RoBERTaMLM, and $Att_{concat,i}$ denotes the attention vector at the i -th time step, L is the time step size.

Leveraging the information from A^{rel} , the attention output is further guided by the emotional polarity of aspect non-opinion words, thereby enhancing the focus on important syntactic keywords.

3.3. SemGCN

The SemGCN module, unlike SynGCN, generates the semantic graph and its corresponding adjacency matrix through AMR [32] instead of utilizing additional syntactic knowledge. By ignoring connectives in sentences and enabling direct associations between aspects and opinion words, AMR provides a more flexible semantic structure compared to traditional approaches. Accordingly, the AMR-extracted representation A^{sem} is employed as the final output of the SemGCN module.

3.3.1. AMR

To better implement the semantic module in ABSA, we use AMR semantic structures as an alternative to traditional semantic relations. First, the AMR graph is parsed from the input sentence using the SPRING parser [33], which generates the AMR representation A^{sem} .

As shown in Fig. 4, after the input sentence “Great food but the service was dreadful.” is processed with the SPRING parser, the AMR relations between the words in the sentence are obtained. For instance, Contrast denotes the contrastive logical relation expressed by “but,” which connects two opposing arguments, ARG1 and ARG2. ARG1 and ARG2 represent the main participants of the predicate. Domain indicates the object of an attribute or adjective. For example, the descriptor “Great” modifies “food”. The numerical suffixes (e.g., 01, 02) represent the automatic indexing mechanism of AMR, ensuring each

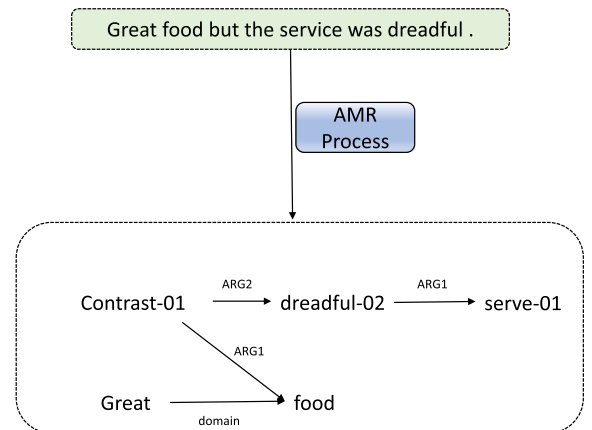


Fig. 4. The SPRING parser processes the sentence.

conceptual node in the graph is unique and traceable. Finally, the reconstructed links between words are derived, resulting in a transformed AMR adjacency matrix where words serve as nodes.

3.3.2. KL-LOSS

To measure the learning discrepancy between SynGCN and SemGCN, we introduce the Kullback-Leibler (KL) divergence. Let $P(X)$ and $Q(X)$ denote the probability distributions obtained by applying non-linear transformations to the representations A^{syn} and A^{sem} from SynGCN and SemGCN, respectively. The KL divergence loss is calculated as follows.

$$l_{kl}(Y) = \sum_{x \in X} P(x) \log \frac{P(x)}{Q(x)} \quad (13)$$

3.4. Auxiliary graph fusion module

To address the varying contributions of different GCN modules across sentences where SynGCN typically captures more information in syntactically rich sentences, we propose an auxiliary graph fusion module inspired by Wang et al. [34]. This module dynamically guides the fusion of syntactic and semantic features, enhancing the discriminative power of the final representation.

3.4.1. Dynamic discrepancy evaluation

To evaluate the reliability of syntactic and semantic features, we introduce dynamic prediction discrepancy as a feedback signal. Unlike static feature fusion, this approach adaptively adjusts fusion weights based on the actual performance of each channel.

For each input sentence, we compute the prediction discrepancies from both syntactic and semantic channels as follows.

$$\begin{aligned} e_{syn} &= \|y - \text{Softmax}(H_{syn}W^1)\|_2 \\ e_{sem} &= \|y - \text{Softmax}(H_{sem}W^2)\|_2 \end{aligned} \quad (14)$$

where W^1 and W^2 denote the weight matrices corresponding to syntactic GCN and semantic GCN, respectively. H_{syn} and H_{sem} represent the input representation vectors to the auxiliary graph fusion module, and y is the one-hot vector of the true label.

To eliminate the impact of error scale variations across different samples, we apply min-max scaling for error normalization:

$$\tilde{e}_{syn} = \frac{e_{syn} - \min(e_{syn})}{\max(e_{syn}) - \min(e_{syn})} \quad (15)$$

$$\tilde{e}_{sem} = \frac{e_{sem} - \min(e_{sem})}{\max(e_{sem}) - \min(e_{sem})} \quad (16)$$

where $\min(\cdot)$ and $\max(\cdot)$ represent the minimum and maximum error values among all samples in the current batch. This normalization scales the L2 norm error term to $[0,1]$, thereby eliminating scale variance.

3.4.2. Adaptive gating mechanism

To effectively fuse syntactic and semantic information, we design an adaptive gating mechanism that dynamically adjusts the fusion ratio between syntactic and semantic information based on input features and error feedback. Unlike traditional weight fusion methods, this mechanism adaptively determines the contribution of syntactic and semantic channels by learning contextual characteristics, thereby enhancing the model's ability to capture complex linguistic structures. The computation process of the adaptive gating mechanism is formulated as follows.

$$G = \sigma([\tilde{e}_{syn} \oplus \tilde{e}_{sem} \oplus H_{syn} \oplus H_{sem}]) \quad (17)$$

where σ denotes the ReLU activation function, and $\oplus$ represents the vector concatenation operation. The output dynamic weight $G \in (0,1)$ modulates the contribution ratio of syntactic channel representation H_{syn} and semantic channel representation H_{sem} in the subsequent fusion process.

3.4.3. Gradient coordination strategy

To achieve effective information exchange between syntactic GCN and semantic GCN, we introduce error gradients as auxiliary features. Specifically, we compute the partial derivatives of the prediction errors with respect to their feature representations for both channels as follows.

$$\nabla e_{syn} = \frac{\partial \tilde{e}_{syn}}{\partial H_{syn}} \quad (18)$$

$$\nabla e_{sem} = \frac{\partial \tilde{e}_{sem}}{\partial H_{sem}} \quad (19)$$

where the gradients reflect the sensitivity of each feature dimension to the error changes. We then construct the auxiliary enhanced features by applying element-wise squaring, which is formulated as follows.

$$H_{au} = \nabla e_{syn} \odot \nabla e_{sem} \quad (20)$$

The final hidden layer representation is computed as follows.

$$H = \alpha H'_{syn} + \beta H'_{sem} + G * H_{au} \quad (21)$$

where H'_{syn} and H'_{sem} represent the feature representations obtained after multiplying H_{syn} and H_{sem} with their corresponding weight matrices W^{syn} and W^{sem} followed by softmax normalization. The hyperparameters α and β control the weights of syntactic and semantic channels, respectively.

3.5. Loss function

To optimize sentiment classification performance, we design a joint optimization loss function. First, we use the standard cross-entropy loss function for ABSA as shown below.

$$l_c(Y) = - \sum_{(s,a) \in D} \sum_{c \in C} \log P(c|s,a) \quad (22)$$

where D contains all sentence-aspect pairs, C represents the set of sentiment polarities, and $P(c|s,a)$ denotes the model's predicted probability that aspect term a in sentence s belongs to sentiment polarity class c .

Finally, we combine the aforementioned KL divergence loss with the cross-entropy loss to form the final loss function.

$$l(Y) = l_{kl}(Y) + l_c(Y) \quad (23)$$

This loss function directly optimizes sentiment classification accuracy while utilizing KL divergence loss to constrain the output distributions of SynGCN and SemGCN, thereby enhancing the synergistic capability of both structural information types. The joint optimization of these two loss components not only improves the model's discriminative power, but also enables more effective fusion of syntactic structures and semantic relations, enhancing the model's generalization capability in ABSA tasks.

4. Experiments

This section presents the experimental setup, including datasets, baseline methods, evaluation metrics, and model parameter configurations, followed by analysis and discussion of comparative experiments and results.

4.1. Datasets

Experiments are conducted on three mainstream public ABSA benchmark datasets. The Twitter dataset [35] consists of tweets collected from Twitter, while the Restaurant and Laptop datasets are from SemEval 2014 Task [36]. Each aspect is annotated with one of three sentiment polarities: positive, neutral, or negative. The statistics of these datasets are presented in Table 1.

Table 1
Statistics for the three experimental datasets.

Dataset	Division	Positive	Negative	Neutral
Rest14	Train	2164	807	637
	Test	727	196	196
Laptop	Train	976	851	455
	Test	337	128	167
Twitter	Train	1507	1528	3016
	Test	172	169	336

4.2. Baseline models

To comprehensively evaluate the performance of our model, we compare it with the following state-of-the-art baseline models.

- TD-LSTM [37]: Extends LSTM to ABSA tasks by employing two LSTM networks for text modeling.
- ATAE-LSTM [4]: Processes hidden layers using LSTM and attention mechanisms, taking both aspect and contextual word embeddings as input.
- IAN [13]: Models the relationship between context and aspect terms using interactive attention, learning representations for both in ABSA tasks.
- MemNet [38]: Utilizes memory networks to predict sentence-based sentiment polarity as aspect sentiment.
- Cabasc [39]: Incorporates word order and relevance into ABSA using two attention enhancement mechanisms.
- AOA [40]: Models aspects and sentences jointly, explicitly capturing interactions between aspects and contextual sentences.
- ASGCN [7]: Constructs graph convolutional networks for ABSA based on syntactic and word dependencies.
- BERT-SPC [41]: Modifies BERT input format to '[CLS] + context + [SEP] + aspect words + [SEP]' for sentence pair classification.
- AEN-BERT [42]: Leverages pre-trained BERT model with attention-based encoders.
- LCF-BERT [42]: Employs local context focus mechanism with context-dynamic masking and weighting layers, combining BERT shared layers to capture both local and global context dependencies.
- R-GAT [43]: Reconstructs dependency trees to remove redundant information and extends the original GNN model with relational attention mechanisms.
- DualGCN [44]: Utilizes dual graph convolutional networks with SynGCN for syntactic structure and SemGCN for semantic associations, incorporating orthogonal and differential regularization.
- SSEGCN [45]: Implements syntax and semantic enhanced graph convolutional networks, combining syntactic and semantic information through an aspect-aware attention mechanism and syntactic masking matrices.
- APARN [9]: Employs AMR path aggregation relation network with path aggregators for semantic feature extraction and relation-enhanced self-attention.
- PHNN [46]: Uses prompt and hybrid neural networks, combining RoBERTa, GCN, and CNN for syntactic features, with BiLSTM and multi-head attention for semantic features.
- DAGCN [34]: Implements distance and aspect-oriented graph convolutional networks with distance-based syntactic weights and aspect fusion attention mechanisms.
- YORO [47]: Adopts a one-to-many framework, simultaneously modeling multiple aspect relationships through globally shared context for sentiment prediction.

- S²GSL [48]: Proposes a method based on fragment-aware semantic graphs and a syntax-based latent graph, and generates an adaptive aggregation network.
- AD-GCFN [26]: Employs semantic and syntactic layers with cross-fusion to integrate semantic and syntactic information.
- KDGCN [27]: Establishes a syntax and semantic GCN architecture, and enhances semantic representation by integrating external commonsense knowledge to mitigate contextual sparsity.

4.3. Evaluation metrics

To evaluate the ABSA models, we employ accuracy and macro-F1 score as the primary evaluation metrics, which are formulated as follows.

$$A = \frac{TP + TN}{TP + FP + TN + FN} \quad (24)$$

$$macro-F1 = \frac{1}{C} \sum_{i=1}^C F1_i \quad (25)$$

where TP , TN , FP , and FN denote true positives, true negatives, false positives, and false negatives, respectively. C is the total number of classes and $F1_i$ represents the $F1$ score for class i .

4.4. Experimental settings

For the word embedding module, we employ the “RoBERTa-base-uncased” version with an embedding dimension of 768 and a RoBERTaMLM embedding dimension of 50,265. In the syntactic GCN module, all syntactic dependency matrices are parsed using the LAL-parser, and the multi-head attention mechanism is configured with 8 heads. The CNN architecture features 1 input channel and 10 output channels. The convolutional kernel size is set to (5, 768) with a stride of (5, 1). The LSTM hidden state dimension is set to 300. For the semantic GCN module, semantic structures are obtained through AMR parsing. All models are configured with a batch size of 16 and a dropout rate of 0.3.

For the restaurant dataset, we set the learning rate to $2e-5$, the regularization coefficient to $1e-2$, train for 50 epochs, with hyperparameters $\alpha = 0.8$ and $\beta = 0.6$.

For the laptop dataset, we configure the learning rate to $2e-5$, the regularization coefficient to $1e-2$, train for 30 epochs, with hyperparameters $\alpha = 0.5$ and $\beta = 0.4$.

For the Twitter dataset, we set the learning rate to $1e-5$, the regularization coefficient to $1e-4$, train for 30 epochs, with hyperparameters $\alpha = 0.6$ and $\beta = 0.6$.

4.5. Main results

Table 2 presents the experimental results of DAGF and baseline models on three datasets, where the best results are shown in bold and the second-best results are underlined. The results of YORO with Twitter are not presented since Twitter’s data format does not suit YORO. The “—” in AD-GCFN and KDGCN indicates that the dataset was not tested. The results show that the DAGF model achieves substantial improvements over state-of-the-art baselines, with accuracy and F1-score improvements of (1.61, 1.92), (1.25, 1.54) and (−0.59, 0.15) on the three datasets, respectively, demonstrating the effectiveness of our approach. Compared to syntax-based models that primarily rely on dependency trees, such as ASGCN and R-GAT, DAGF demonstrates significant improvements, achieving an average accuracy increase of 5.5% across the three datasets. While syntax-based models can effectively capture structural dependencies, they perform poorly when processing sentences that contain weak syntactic signals or implicit semantic relations. For example, on the Twitter dataset, where informal expressions are prevalent, ASGCN’s performance is noticeably limited, whereas DAGF

Table 2

Experimental results obtained on the Restaurant, Laptop, and Twitter datasets. Acc denotes accuracy, and F1 represents the macro F1-score. The best results are shown in bold, and the second-best results are underlined.

Models	Restaurant		Laptop		Twitter	
	Acc	F1	Acc	F1	Acc	F1
TD-LSTM [37]	78.66	66.90	69.44	63.06	70.52	69.25
ATAE-LSTM [4]	77.77	65.92	67.75	61.91	67.34	65.27
IAN [13]	77.32	64.55	68.81	62.12	67.77	65.07
MemNet [38]	77.95	65.51	70.38	64.67	70.23	68.69
Cabasc [39]	77.68	64.67	69.12	64.33	67.63	64.23
AOA [40]	79.46	66.68	73.82	67.84	70.95	69.18
ASGCN [7]	79.82	70.67	74.92	70.92	70.66	68.28
BERT-SPC [41]	85.18	77.27	78.68	75.22	75.14	73.15
AEN-BERT [42]	80.89	70.71	77.74	73.30	73.27	71.43
LCF-BERT [42]	86.61	79.57	79.47	75.56	73.41	70.66
R-GAT [43]	86.60	81.35	78.21	74.07	76.15	74.88
DualGCN [44]	86.33	80.51	81.20	78.10	76.36	75.49
SSEGCN [45]	87.76	82.45	80.06	76.59	77.84	76.40
APARN [9]	87.31	81.79	79.91	76.21	77.13	75.93
PHNN [46]	87.59	81.37	81.66	78.84	75.29	73.95
DAGCN [34]	85.88	80.02	81.80	78.64	75.78	74.43
YORO [47]	87.14	81.83	<u>81.82</u>	78.32	—	—
S ² GSL [48]	87.31	82.84	82.46	79.07	76.87	76.29
AD-GCFN [26]	—	—	77.90	74.14	75.00	74.03
KDGCN [27]	85.31	79.31	79.91	76.84	—	—
Ours	89.37	84.37	83.07	80.38	<u>77.25</u>	76.55

incorporates AMR-based semantic information, thus compensating for these syntactic ambiguities.

Compared to semantic-based methods, such as APARN, which focus on semantic structures, DAGF exhibits consistent superiority, especially on the Laptop dataset where sentences tend to be longer and more complex. Pure semantic models struggle to capture long-range dependencies explicitly represented in the syntactic structure. DAGF addresses this limitation by integrating dependency tree information via a syntactic GCN, leading to more accurate aspect-opinion pairing in structurally complex sentences.

Compared to existing dual-graph architectures, including DualGCN, SSEGCN, S²GSL, AD-GCFN, and KDGCN, DAGF achieves superior performance on the test datasets. Specifically, DAGF's accuracy is higher than these baseline models by an average of 2.65%, indicating that our auxiliary graph fusion module integrates syntactic and semantic information more effectively than existing fusion strategies. The key advantage lies in our dynamic fusion mechanism, which can adaptively adjust the weight distribution between syntactic and semantic features based on sentence characteristics, rather than treating them as separate channels with a fixed combination ratio.

DAGF's performance is particularly outstanding on the Restaurant dataset, with an accuracy improvement of 1.61 and an F1-score improvement of 1.92 compared to the best baseline model. This can be attributed to the more formal expressions and clearer syntactic structures often present in restaurant reviews, allowing both components of the GCN to function optimally. Notably, on the Twitter dataset, which is characterized by informal and syntactically ambiguous language, DAGF achieves a highly competitive second-best accuracy (77.25), only slightly behind SSEGCN (77.84), while attaining the best F1-score (76.55) among all compared models. This demonstrates DAGF's robustness in scenarios where pure syntactic parsing is less reliable, as it effectively leverages semantic information to compensate for informal expressions. The performance variations across different datasets further validate the effectiveness of our model in ABSA tasks.

4.6. Ablation study

In this section, we conducted ablation experiments on DAGF across the restaurant, laptop, and Twitter datasets to validate the performance of each module in our method. The experimental results are presented

Table 3

Ablation analysis of DAGF model.

Models	Restaurant		Laptop		Twitter	
	Accuracy	Macro-F1	Accuracy	Macro-F1	Accuracy	Macro-F1
Ours	89.37	84.37	83.07	80.38	77.25	76.55
w/o Lexicon	88.36	81.77	81.65	79.53	74.33	73.88
w/o CNN	88.42	83.65	80.74	79.39	76.43	72.98
w/o LSTM	87.94	82.78	81.33	79.84	75.16	73.03
w/o AMR	87.74	83.65	82.28	80.22	76.58	73.26
w/o AGF	87.55	82.54	82.08	81.66	74.66	73.87

in Table 3, where “w/o Lexicon” indicates that only the syntactic dependency tree is used in the syntax module. “w/o AMR” means that AMR is not used to represent semantic structures. “w/o AGF” indicates that the fusion of syntactic and semantic modules is done merely through concatenation, showing that removing any module leads to a decline in model performance. For instance, after removing the opinion lexicon, the model's accuracy and F1 scores decrease by (1.01, 2.6), (1.42, 0.85), and (2.92, 2.76) on the three datasets, respectively, demonstrating that establishing connections between the opinion lexicon and non-opinion words enhances performance. After removing the CNN and LSTM modules, both accuracy and F1 scores decreased across the three datasets, indicating that CNN and LSTM effectively extract syntactic information from the merged data. We also observe that removing the AMR semantic structure has a relatively minor impact on final performance for the Twitter dataset, while showing a more significant effect on the restaurant and laptop datasets. This indicates that our DAGF model relies more heavily on semantic information from sentences when processing the restaurant and laptop datasets. Furthermore, we find that removing the auxiliary graph fusion module has a greater impact on the restaurant and laptop datasets. On the Twitter dataset, eliminating the auxiliary graph fusion module results in decreases of 2.69 and 2.68 in overall model accuracy and F1 scores, respectively, as this module significantly improves the fusion efficiency of syntactic and semantic information.

4.7. Case study

To further investigate the ABSA effectiveness of different models, we collected nine aspect terms from five sentences in the test set. As shown in Table 4, AEN-BERT, BERT-SPC, ATAE-LSTM, and ASGCN are compared with DAGF for analysis. The symbols P, O, and N represent positive, neutral, and negative sentiments, respectively. The “Target” column shows the true sentiments of sentences. Additionally, we bold the results of the DAGF model to clearly display the prediction outcomes.

For the aspect “portions of the food” in the first example, the attention-based method (ATAE-LSTM) ignores the syntactic information of the sentence and incorrectly treats “food” as the aspect, associating it with the opinion word “mediocre.” In the second example, “Bartender” and “drinks” are syntactically distant from each other. Therefore, syntax-based models (AEN-BERT and ASGCN) perform poorly. In the third example, the non-opinion word “fresh” belongs to positive words in the opinion lexicon. Thus, our model associates the aspect term “fresh juice concoctions” with the non-opinion word “fresh,” thereby better analyzing the sentiment polarity. This demonstrates that non-opinion words are also an aspect that should not be overlooked when analyzing sentiment polarity.

4.8. Hyperparameter learning rate study

To determine the optimal values of the fusion coefficients α and β , we employed the Optuna hyperparameter optimization framework [49], which is based on Bayesian optimization. Specifically, the Tree-structured Parzen Estimator (TPE) [50] sampler was used to efficiently explore the parameter space. The search ranges were set to $\alpha \in [0.0, 1.0]$ and $\beta \in [0.0, 1.0]$, with accuracy used as the objective function. We

Table 4
Results of case studies conducted on three datasets.

Sentence	Aspect	AEN-BERT	BERT-SPC	ATAE-LSTM	ASGCN	DAGF	Target
The portions of the food that came out were mediocre	Portions of the food	O	N	N	N	O	O
Well, it happened because of a graceless manager and a rude bartender who had us waiting 20 minutes for drinks, and then tells us to chill out	Bartender Drinks	N N	O O	O O	N N	N O	N O
Be sure to accompany your food with one of their fresh juice concoctions	Food Fresh juice concoctions	P P	O P	P P	P P	O P	O P
From the speed to the multi touch gestures this operating system beats Windows easily	Speed	P	P	P	P	N	N
	Windows	P	P	P	P	P	P

Table 5
Results of hyperparameter tuning for α and β using Optuna.

α	β	Restaurant Acc	Laptop Acc	Twitter Acc
0.3	0.7	85.62	81.49	74.33
0.6	0.6	86.11	80.73	77.25
0.5	0.4	87.03	83.07	75.89
0.7	0.3	87.26	79.64	73.49
0.8	0.6	89.37	81.01	75.62
0.9	0.8	86.89	80.94	72.58
0.4	0.9	85.94	81.26	73.68
0.2	0.5	85.77	78.63	71.80

conducted 50 independent trials, and a median pruning strategy was applied to terminate unpromising trials early.

To avoid redundancy, Table 5 reports only eight representative trials that illustrate the overall performance trends across different parameter combinations.

From the above experimental results, we observe that for the restaurant and laptop datasets, increasing α (decreasing α) and decreasing β (increasing β) both significantly affect model performance, indicating an imbalance in the importance of syntactic and semantic information in sentences from the restaurant and laptop datasets. Balancing their weight proportions can effectively improve model performance. In the Twitter dataset, increasing α (decreasing α) and decreasing β (increasing β) do not have a substantial impact on model performance. The main issue is that the Twitter dataset contains numerous informal expressions, leading to lower contributions of syntactic and semantic features to the ABSA task, which prevents the model from extracting effective corpus information.

4.9. Statistical significance analysis

To rigorously validate the effectiveness of our proposed model, we conducted comprehensive statistical significance tests using paired t-tests with a significance level of $\alpha = 0.05$. We selected five competitive

baseline models that achieved comparable performance: SSEGCN [45], PHNN [46], APARN [9], YORO [47] and DualGCN [44]. Table 6 presents the detailed statistical significance test results across all three datasets.

On the Restaurant dataset, our model demonstrates statistically significant improvements over all five baselines with $p < 0.01$ for all comparisons, with accuracy gains ranging from 1.61 to 3.04. The Laptop dataset shows even stronger results, where all comparisons yield $p < 0.01$ and improvements range from 1.25 to 3.16, with three comparisons reaching $p < 0.001$. The Twitter dataset presents more modest improvements due to its noisy and informal nature, yet our model still achieves statistically significant gains in 3 out of 4 comparisons. Notably, while SSEGCN achieves slightly higher accuracy, the statistically significant difference indicates consistent performance variation rather than random fluctuation. The comparison with APARN shows no significant difference, suggesting similar performance on this challenging dataset.

In summary, the proposed model demonstrates statistically significant improvements across all datasets. The consistently low p-values, especially in the Restaurant and Laptop datasets, provide robust statistical evidence that the observed performance gains are both reliable and reproducible.

4.10. The semantic GCN layer number study

In this section, we investigated the impact of GCN layers ranging from 1 to 6 on the semantic component across the restaurant, laptop, and Twitter datasets. As shown in Fig. 5, experimental results demonstrate that the semantic component achieves optimal performance when using a 1-layer GCN. As the number of layers increases, accuracy fluctuates. This occurs because when the GCN has 1 layer, each node primarily focuses on its directly adjacent nodes, and these neighboring nodes typically contain more direct sentiment-related information. As the number of layers increases, node representations gradually become similar. This makes it difficult for the model to distinguish between different nodes and aspect sentiments, resulting in more redundant information

Table 6
Statistical significance test results comparing our model with competitive baselines. The paired t-test is conducted on all three datasets with significance level of $\alpha = 0.05$. Δ Acc denotes the accuracy improvement of our model over the baseline. * indicates $p < 0.05$, ** indicates $p < 0.01$, and *** indicates $p < 0.001$.

Model	Restaurant			Laptop			Twitter		
	Acc	Δ Acc	p-value	Acc	Δ Acc	p-value	Acc	Δ Acc	p-value
Ours	89.37	—	—	83.07	—	—	77.25	—	—
SSEGCN	87.76	+1.61	0.0023**	80.06	+3.01	<0.001***	77.84	-0.59	0.0325*
PHNN	87.59	+1.78	0.0011**	81.66	+1.41	0.0045**	75.29	+1.96	0.0008***
APARN	87.31	+2.06	0.0005***	79.91	+3.16	<0.001***	77.13	+0.12	0.6251
YORO	87.14	+2.23	0.0003***	81.82	+1.25	0.0089**	—	—	—
DualGCN	86.33	+3.04	<0.001***	81.20	+1.87	0.0013**	76.36	+0.89	0.0267*

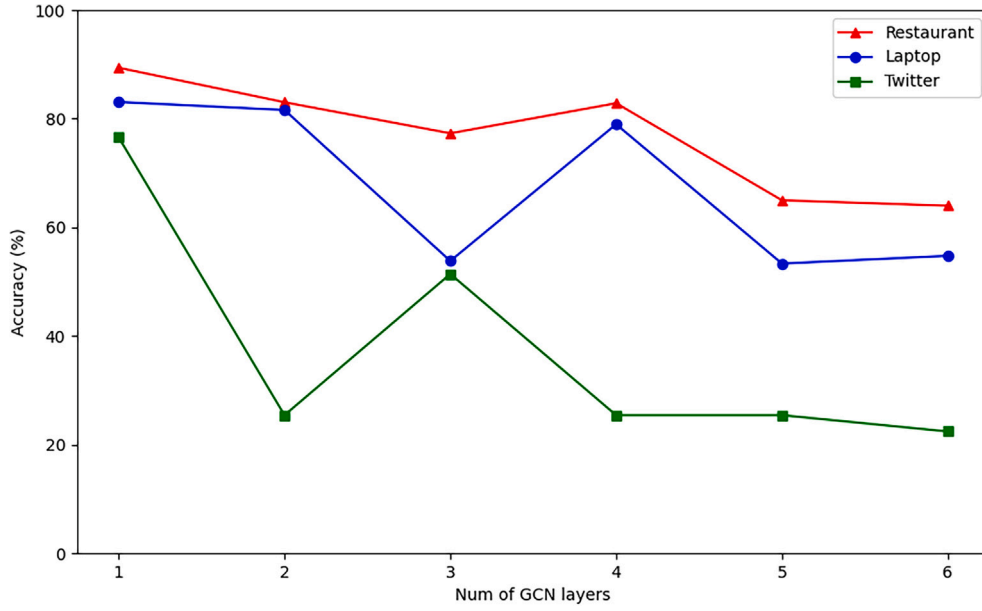


Fig. 5. The influence of the number of GCN layers on the semantic part.

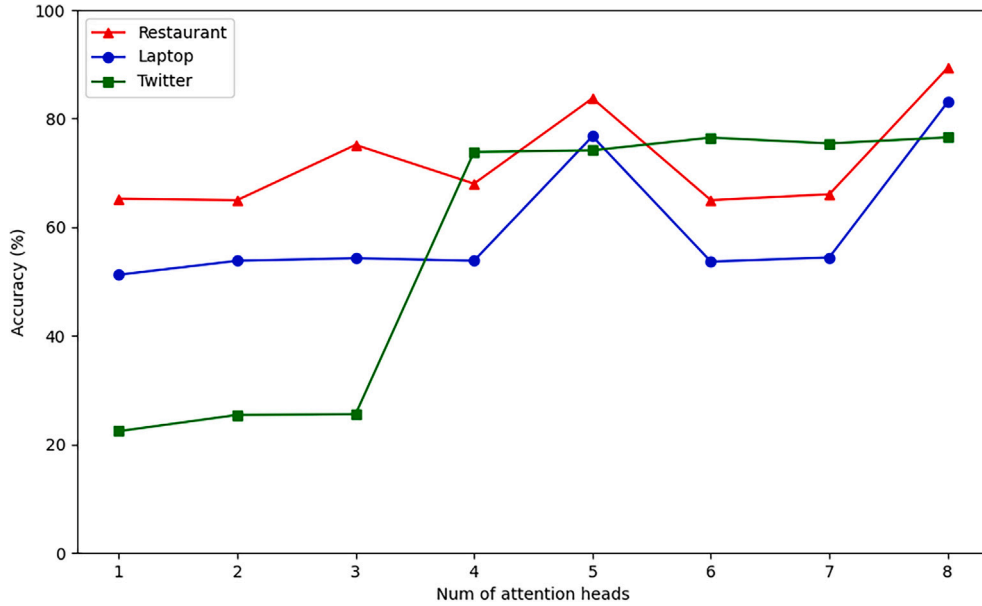


Fig. 6. The influence of the number of attention heads on the syntactic part.

and poorer model performance. In summary, when processing semantic information, the 1-layer GCN structure performs best in our model. This indicates that directly adjacent information is more critical for aspect sentiment analysis, while deeper layers introduce noise instead. Therefore, in the final model proposed in this paper, we adopt a 1-layer GCN for feature extraction in the semantic component.

4.11. Relation-aware self-attention mechanism study

In this section, we investigate the impact of the number of relation-aware self-attention heads in the syntactic component on model performance. As shown in Fig. 6, experimental results demonstrate that the model achieves optimal performance when the multi-head attention has 8 heads. First, when the multi-head attention count is set to 1, the model can only focus on a single syntactic relation. This makes the model overly constrained when learning syntactic dependencies and limits its ability to capture complex grammatical structures. Second, when

increasing the number of attention heads, the model can learn multiple different syntactic relations in parallel, fully exploiting different syntactic patterns. For example, some heads can focus on word order structures while others can attend to long-range dependencies in sentences. For the relation-aware self-attention mechanism in the syntactic component, using 8 attention heads can most effectively capture complex syntactic dependencies, thereby achieving optimal model performance. Therefore, in the final model architecture proposed in this study, the relation-aware self-attention mechanism in the syntactic component is configured to use 8 heads.

5. Conclusion and future work

This paper proposes a dual GCN and auxiliary graph fusion-based ABSA model called DAGF, which simultaneously considers both syntactic and semantic information of sentences. By fusing the opinion lexicon with syntactic dependency trees, DAGF enhances the extraction

of syntactic feature information from sentences. Meanwhile, it uses AMR to construct semantics-based adjacency matrices. Finally, through the auxiliary graph fusion module, it adjusts the proportion of syntactic and semantic information. This effectively distributes the weights of sentiment information and ensures the completeness of sentence information. Experimental results on publicly available datasets demonstrate that the DAGF model can effectively improve performance metrics for sentiment polarity classification. While the DAGF model demonstrates strong performance by effectively integrating syntactic and semantic graphs, its design entails certain computational trade-offs. The reliance on AMR introduces preprocessing overhead, and the dual-GCN architecture increases memory consumption during training. Furthermore, the adaptive fusion mechanism introduces parameter sensitivity that requires tuning for new domains. Future work will focus on developing more efficient, end-to-end parsers and exploring distilled or lightweight graph learning techniques to mitigate these costs while preserving the model's representational strengths. Additionally, we plan to explore synergies by integrating knowledge from large language models (LLMs) to enhance semantic understanding and investigating efficient hybrid architectures that combine our graph-based method with state-space models (SSMs) for long-sequence processing.

CRedit authorship contribution statement

Jie Ji: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Wenlong Zhu:** Writing – review & editing, Validation, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Chengle Hou:** Writing – review & editing, Validation, Software, Investigation. **Qiaoyan Song:** Validation, Software. **Yukun Ma:** Validation. **Youruo Wang:** Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data is available.

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